

Course: Web APIs
Course Credits: 4

Course Code:
Learning hours: 120

Course Outcomes

CO1: Familiarize with all the features of Spring Boot
CO2: Learn to query data in Spring Boot.
CO3: Design and develop RESTful web services with Spring Boot.
CO4: Develop microservices with Spring Boot.
CO5: Build Spring applications using Spring Boot.

Unit 1: Spring Boot – Introduction

Introduction to Spring Boot; Need for Spring Boot; Spring Boot Features; Spring vs Spring Boot vs Spring MVC; Creating Project- Spring Initializr; Spring Boot Hello World Application.

***Learning Outcome:** Gain an introduction to Spring Boot.*

Unit 2: Properties, Annotations and Dependencies

Command Line Properties, Properties File, YAML File, Externalized Properties, @Value Annotation, Spring Boot Active Profile, @SpringBootApplication; Spring Boot Annotations; Dependency Management – Maven Dependency, Gradle Dependency

***Learning Outcome:** Understand Spring Boot's properties, annotations, and dependencies.*

Unit 3: Actuator, Code Structure, Configuration Classes

Actuator – Overview, Capabilities, Actuator Endpoints, Custom Actuators;
Spring Boot Code Structure – Default Package, Typical Layout;
Importing Additional Configuration Classes; Importing XML Configuration;
Spring Beans and Dependency Injection

***Learning Outcome:** Understand Spring Boot's actuator, code structure and configuration classes.*

Unit 4: Spring Boot Tomcat Custom Port Number and Spring Boot Rest Template

Spring Boot Port Number – Custom Port, Random Port; Spring Boot Rest Template – Introduction to REST, SOAP vs REST, GET, POST, PUT, DELETE

***Learning Outcome:** Understand Spring Boot's Port Number and REST template*

Unit 5: Spring Boot Database Handling

Spring Boot JPA CRUD Application; JDBCTemplate; Multiple DataSource; CRUD

***Learning Outcome:** Learn about Spring Boot's Database handling*

Unit 6: Working with NoSQL MongoDB

Connecting to MongoDB Database; MongoTemplate; Spring Data MongoDB Repositories; Embedded Mongo

Learning Outcome: Learn about Spring Boot's Database handling

Unit 7: Building RESTful Web Services

Rest Controller; Request Mapping; Request Body; Path Variable; Request Parameter; GET API; POST API; PUT API; DELETE API; Introduction to RESTful Web Service; Initializing RESTful Web Services

Learning Outcome: Learn about Spring Boot's Database handling

Unit 8: Spring Boot with Microservices

Microservices – Introduction, Architecture, Project Structure; Spring Boot with Microservices Basics; Coding Standards; Builder Design Pattern

Learning Outcome: Learn about Spring Boot Microservices

Unit 9: Deploy Spring Boot Applications on External Tomcat Server

Spring Boot Servlet Initializer; Setting Main Class; Updating packaging JAR into WAR; Packaging Application; Deployment in Tomcat

Learning Outcome: Deploy Spring Boot Applications on Tomcat Server

Unit 10: Spring Boot Payloads – XML and JSON

Enable CORS in Controller Method; Global CORS Configuration; Spring Boot with XML Payload; Spring Boot with JSON Payload

Learning Outcome: Learn about Spring Boot Payloads

Unit 11: Developing Web Applications – Spring Web MVC Framework

Spring MVC Auto-Configuration; HttpMessageConverters; MessageCodesResolver; Static Content; Path Matching and Content Navigation; ConfigurableWebBindingInitializer; Spring HATEOS; CORS Support

Learning Outcome: Use Spring Web for developing a web application.

Unit 12: Developing Web Applications – Spring WebFlux MVC Framework

Spring WebFlux Auto-configuration; HTTP Codecs with HTTPMessageReaders and HttpMessageWriters; Static Content, Web Filters

Learning Outcome: Use Spring WebFlux for developing a web application.

Unit 13: Spring Boot View

What is Thymeleaf?; Need of Thymeleaf; Features; Template Types – Thymeleaf Support; Standard Dialect

***Learning Outcome:** Learn to use the ThymeLeaf Template Engine*

Unit 14: Spring Boot AOP, Runners, and Log

AOP Before Advice, After Advice, Around Advice, Returning Advice, Throwing Advice; Application Runner; Command-Line Runner; Logging – Log Format, Console Log Output, File Log Output, Log Levels, Configure Logback

***Learning Outcome:** Gain an introduction to aspect oriented programming.*

Unit 15: Spring Boot Security

Securing Web Applications; Spring Boot Security – Authorization Server, Resource Server, OAuth2, JWT Token

***Learning Outcome:** Familiarize with the security aspects of Sprint Boot*